

A095 – Inside/Outside Topology of the Torus and Chiral Projection

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.1 Purpose

[STIPULATION] This document closes the gap between half-integrality on the torus (A263) and chiral projection into flat space (A090, A264). The core: a torus has an inside and an outside with topologically different coupling directions. After decompactification these appear as two chiral states – and the projection is not symmetric.

.2 Half-Integrality and Torus Geometry

[B] Spin- $\frac{1}{2}$ follows from the winding $(n_\theta, n_\phi) = (1, 1)$ on T^4 (A263): two rotations (720°) bring the spinor back to its initial state. This is pure topology.

[B] A torus with winding $(1, 1)$ has an intrinsic distinction: inside and outside. For an embedded torus in $\mathbb{R}^3$: the Gaussian curvature is positive on the outside ($K > 0$) and negative on the inside ($K < 0$). For the flat identification torus T^4 , $K = 0$ everywhere – but the poloidal/toroidal circulation structure nevertheless defines a topological inside/outside distinction through the orientation of the winding.

[B] The field direction is topologically fixed by the winding geometry: the electric field vector points inward on the outside (for negative charge) and outward (for positive charge). This direction is not freely choosable – it follows from the quantised winding number $\Phi = n \cdot h/e$.

.3 Chiral Projection

[B] The decompactification $T^4 \rightarrow \mathbb{R}^3 \times \mathbb{R}_t$ (type-I projection, A090) is not symmetric with respect to inside and outside. The compact torus carries both sides simultaneously; the decompactified flat space sees only one projection.

[B] What remains after projection:

- The *outer state* projects to left-handed coupling (positive rotation phase $+\xi/2$).

- The *inner state* projects to right-handed coupling (negative rotation phase $-\xi/2$).

The projection selects one side – this is the geometric origin of chiral asymmetry.

[B] Why this explains the chiral structure. In the compact T^4 , both sides couple with equal strength – there is no preferred direction. After decompactification, only the outer perspective exists as an observable state. The inner state is not directly accessible through the type-I projection – it is the “other branch” that is lost in the dimensional reduction (A090, type-II loss).

.4 Connection to the Weak Interaction

[B] Fourier projector and $g_R = 0$. The Hilbert space on T^4 decomposes into sectors by the winding index m . The physical projector P_+ selects the $m \geq 0$ sector (outer state):

$$P_+ = \sum_{n, m \geq 0} |n, m\rangle \langle n, m|.$$

Action on the two chiral states:

$$P_+|1, +1\rangle = |1, +1\rangle \quad (\text{outer state survives}), \quad P_+|1, -1\rangle = 0 \quad (\text{inner state lies in the kernel}).$$

After projection the inner state is $|0\rangle$ – a null vector. The weak coupling:

$$g_L = \langle 1, +1 | H_{\text{weak}} | 1, +1 \rangle \neq 0, \quad g_R = \langle 0 | H_{\text{weak}} | 0 \rangle = 0.$$

This is not a sign argument ($g_R = -g_L$) but a kernel argument: the inner state lies in the kernel of P_+ and has no physical existence after projection. $g_R = 0$ follows algebraically.

[S] What remains open: why P_+ and not P_- ? The projector P_+ corresponds to the outer side ($K > 0$, positive orientation, electric flux inward). P_- would be the inner-state projector. That P_+ is the physical projector follows from orientation arguments (Gaussian curvature, field rotation $\nabla \times \mathbf{E}_{\text{field}}$) – but not from a derivable symmetry breaking. This is the one remaining open edge.

.5 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$(n_\theta, n_\phi) = (1, 1)$	Winding numbers for spin- $\frac{1}{2}$ on T^4
K	Gaussian curvature; outside $K > 0$, inside $K < 0$
$\Phi = n \cdot h/e$	Quantised flux lines, topologically fixed field direction
g_L, g_R	Left- and right-handed weak coupling
Type-I projection	Information-preserving decompactification (A090)
Type-II loss	Non-reversible spatial projection, information loss

.6 Verification Script

- Topological consistency: winding (1, 1), 720° symmetry, Gaussian curvature inside/outside, phase difference $\pm\xi/2$: `python/A_Serie_Skripte/a095_torus_chiralitaet.py`

.7 Open Questions

$g_R = 0$ is now at [B] level: given P_+ as the physical projector, $g_R = 0$ follows algebraically from $P_+|1, -1\rangle = 0$.

What remains [S]: why P_+ is the physical projector. The orientation argument (Gaussian curvature, field rotation) is geometrically plausible, but not derived from a deeper symmetry breaking. This is the remaining open edge – and it is precisely named.

Connection to the gauge sector (A190) missing. The chiral projection explains the $SU(2)_L$ coupling qualitatively; whether it reproduces the full electroweak gauge structure is not shown.

.8 Supersedes

This document systematises the inside/outside chirality mechanism from the legacy corpus (particle mass torus model, Dirac winding structure, projection chain framework). It supplements A263 (Dirac, half-integrality), A090 (projection chain, type-I/II), and A264 (asymmetry, CP) with the missing link.

.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.